



Exosome miR-152-3p derived from small intestinal epithelium modulates aging process in adipocytes

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Abstract

Exosomes play a crucial role in facilitating intracellular communication between cells and tissues. The small intestine epithelium secretes exosomes, which is involved in various physiologic and pathologic processes. In this study, we investigated the effects of exosomal miR-152-3p derived from small intestinal epithelium on the aging process of adipocytes and its potential downstream mechanism. The exosomes derived from small intestinal epithelial cells were identified and characterized by TEM, NTA, and Western blot (WB). CCK-8 assay demonstrated the concentration-dependently increased 3T3-L1 cell viability by exosomes. PCR, Mito-Tracker red and DCFH-DA staining demonstrated the increased mtDNA content, mitochondrial activity, and the declined ROS content in 3T3-L1 adipocytes co-cultured with young exosomes. WB, PCR, β -galactosidase staining and ELISA demonstrated that the senescence was suppressed, uncoupling protein 1 (UCP1) and PPARgamma coactivator 1-alpha (PGC-1 α) expression were upregulated, the levels of proinflammatory tumor necrosis factor-alpha (TNF- α) and interleukin 6 (IL-6) were decreased in 3T3-L1 adipocytes co-cultured with young exosomes. Luciferase reporter assay determined the binding between miR-152-3p and PGC-1 α . WB and PCR manifested that miR-152-3p was lowly expressed in young exosomes and miR-152-3p could decrease PGC-1 α expression and increase the expression of senescence-related genes. Moreover, ITT and GTT and H&E staining in *in vivo* elderly mouse model demonstrated that miR-152-3p inhibitor decreased visceral fat, improved glucose tolerance and insulin sensitivity and inhibited aging. WB and PCR suggested that miR-152-3p inhibitor enhanced PGC-1 α expression, suppressed the expression of senescence-related genes and proinflammatory factors *in vivo*. In summary, intestinal exosomes affect the browning function of adipocytes through miR-152-3p, modulating the aging process.

Keywords Small intestinal epithelium · Exosomes · MiR-152-3p · Senescence

Introduction

Aging is a normal physiologic process accompanied by cognitive decline. The aging process is associated with an increased incidence of disease conditions (Kopp 2024). The accumulation and dysfunction of adipose tissues are early aging characteristics in humans (Munoz-Espin and Serrano 2014; Kimura et al. xxxx). Adipose tissue is the

largest energy storage and endocrine organ, traditionally divided into white adipose tissue (WAT) and brown adipose tissue (BAT), which play an important role in energy and metabolism (Liu et al. 2013). Mice designed to have high levels of brown and/or beige fat activity have been found to resist weight gain and exhibit healthy metabolic characteristics (Kopecky et al. 1995; Guerra et al. 2025). Studies have found that dysfunction of adipose tissues during aging promotes mild chronic inflammation, insulin resistance, and lipid infiltration in the elderly, while the increased activity of brown fat may promote beneficial metabolic effects (Cypess et al. 2015; Park et al. 2014; Ghosh et al. 2015; Camell et al. 2017). However, the mechanisms underlying inter-organ interactions and the aging process are still poorly understood, and the potential mechanisms for addressing aging issues need to be further explored.

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Exosomes are extracellular vesicles of a diameter of 50–100 nm, which are formed by multivesicular endosomes and released by almost every cell type (Zhang et al. 2020). Communication between cells is facilitated by exosomes that deliver mRNA, microRNA, proteins, and signaling molecules (Liang et al. 2019). Van Niel et al (Niel et al. 2001). found that intestinal epithelial cells can secrete extracellular vesicles from their top and bottom outer sides. The small intestine epithelium secretes extracellular vesicles carrying MHC class II/peptide complexes, which are involved in activating the immune system (Niel et al. 2003). As an important organ, the intestine primarily digests and absorbs nutrients to maintain energy homeostasis, and has also been found to be involved in the regulation of fat metabolism (Everard et al. 2014; Backhed et al. 2004). However, few studies have addressed the regulatory mechanism of exosomes derived from the small intestine (Yang et al. 2024).

MicroRNA (miRNA) is a non-protein-encoded microRNA, which plays a crucial role in various pathologic conditions and developmental processes. Previous study has found that the upregulation of miR-30a-5p, as well as the downregulation of miR-30a-3p and miR-181a-5p, can stimulate the induction of aging in vascular endothelial cells (Toyama et al. 2022). Ectopic overexpression of microRNA-34a (miR-34a) can induce cell cycle arrest, apoptosis, and aging (Boon et al. 2013; Liu et al. 2011). In this study, we selected miR-152-3p as the target to investigate its effect on adipocytes.

We demonstrated that exosomes in the small intestine of young mice enhanced the viability of adipocytes and that exosomes might affect the molecular mechanism of adipocyte differentiation through miR-152-3p. Therefore, targeting miR-152-3p and/or small intestinal exosomes might be an attractive strategy for retarding aging and related diseases.

Materials and methods

Animal studies

Adult male C57BL/6 J mice (8 weeks or 18 months old) were obtained from the Model Animal Research Center of Nanjing University (Nanjing, China). Mice were maintained in a 12 h light/dark cycle (with lights on at 7 a.m.) and housed in conditions of 23 ± 1 °C and 60% humidity, with free access to food and water.

For the glucose tolerance test (GTT), the mice kept on fasting for 16 h before intraperitoneal (IP) injection with D-glucose (2 g/kg). For the insulin tolerance test (ITT), the mice have fasted for 6 h followed by intraperitoneal injection of human insulin (0.75 units/kg).

Histologic analysis

Fat around the kidney tissues from the mice was collected, immersed in 10% formalin solution (Macklin, China), and fixed overnight. Following dehydration in gradient ethanol, transparency in xylene, and embedment in paraffin, 4- μ m-thick tissue sections were prepared. For histologic analysis, the sections were dewaxed and rehydrated before staining.

For hematoxylin and eosin (H&E) staining, sections were sequentially treated with hematoxylin and eosin (Phygene, China). Subsequently, the excess staining solution was removed before observation under an optical microscope.

Isolation and culture of the small intestinal epithelial cells (IECs)

The young or old mouse gut was dissected under a magnifying glass to remove the mesenteric tissue associated with the intestines. Tissues were washed with PBS and diced into 1 mm³ pieces, then digested for 37 min at 25 °C using a digestion buffer containing Collagenase type I and hyaluronidase. The supernatant was obtained by centrifugation after the mixture had settled for 5 min. After centrifugation at 100 g for 5 min, supernatants were removed and cells were resuspended in DMEM (Gibco, USA). The cells were cultured on petri dishes coated with poly-L-lysine (Sigma, USA).

Isolation and identification of exosomes from IECs (IECs-exos)

To extract the exosomes, the medium was collected and centrifuged (10 min, 500 g; 30 min, 3500 g; 60 min, 10,000 g; 120,000 g, 3 h). The morphology of precipitated particles was observed and the morphology of exosomes was identified using transmission electron microscopy (TEM, Hitachi HT7700, Tokyo, Japan). Western blot (WB) was used to detect the exosome-specific biomarkers CD9, CD63, tumor susceptibility gene 101 (TSG101), and GM130. Exosomes were lysed in RIPA buffer and total protein concentration was determined by BCA (Bioteke, China). WB analysis was performed using the following antibodies: CD9 antibody (ab307085, dilution ratio of 1:2000), CD63 antibody (ab134045, dilution ratio of 1:2000), TSG101 (ab125011, dilution ratio of 1:2000), and GM130 (PA5-95,727, dilution ratio of 1:1000). Exosome diameters and exosome numbers were determined using NTA technology (ZetaView, Particle Metrix, Germany). The NTA measurement conditions were: temperature = 23.4–23.6; viscosity = 0.919–0.921 cP; frames per second = 25; measurement time = 91 s.

Cell treatment and transfection

Murine 3 T3-L1 cells (obtained from American Type Culture Collection) were grown to confluence and differentiated into adipocyte (Lu et al. 2005). A culture medium containing DMEM and 10% fetal bovine serum was used to cultivate preadipocytes. The cells were differentiated in DMEM containing 10% fetal bovine serum for 2 days by adding 1 µg/mL insulin, 0.5 mM isobutylmethylxanthine (IBMX), and 1 µM dexamethasone. After that, cells were maintained in DMEM supplemented with 10% fetal bovine serum. The experiments were performed on day 10 and 12 post-differentiation.

The mimic negative control (mimic NC), miR-152-3p mimics, miR-152-3p inhibitor, or inhibitor negative control (inhibitor NC) were synthesized by Ribobio's (Guangzhou, China). After being seeded into 24-well plates for 24 h, 3 T3-L1 cells were respectively transfected with mimic NC, miR-152-3p mimics, inhibitor NC or miR-152-3p inhibitor by lipofectamine 3000 (50 nM; Invitrogen, USA).

Oil red O staining

After PBS washing, the matured 3 T3-L1 cells were fixed with 10% buffered formalin for 30 min at room temperature, washed with isopropanol (60%), followed by incubation with Oil-Red-O working solution for 1 h (Kim et al. 2025). The stained lipid droplets were captured using a light microscope at ×40 magnification.

Co-culture of intestinal exosomes with 3 T3-L1 cells

To investigate the role of exosomes, 3 T3-L1 cells were incubated with exosomes at the concentrations of 0, 1×10^5 , 1×10^6 , 1×10^7 , 1×10^8 and 1×10^9 particles/mL for 24 h. The proliferation of cells was determined using the CCK8 assay kit according to the manufacturer's instructions (Beyotime, China).

Analysis of mitochondrial content using MitoTracker red

3 T3-L1 cells were seeded and received indicated treatment in the control, young-exosomes, and old-exosomes groups separately. 3 T3-L1 cells were incubated with exosomes at the concentrations of 10^7 particles/mL. Mitochondria were labeled using the MitoTracker Red CMXRos (Beyotime, China) according to the manufacturer's protocol. Cell nuclei were stained with 4',6-diamidino-2-phenylindole (DAPI, Beyotime, China). The cells were observed under a

fluorescent microscope at ×20 magnification and the fluorescence intensity was measured with ImageJ software.

Real-time PCR analysis to detect mitochondrial DNA copy number

To detect cellular mitochondrial DNA (mtDNA) copy number, quantitative real-time PCR was used. DNA primers were designed to detect Complex II (succinate-ubiquinone oxidoreductase), as markers for mitochondrial DNA (mtDNA) (Guo et al. 2014). The following primers were used: mtDNA: F-CAAACCTACGCCAAAATCC, R-GAAATGAATGAGCCTACAG; and GAPDH: F-TGACAACAGCCTCAAGA, R-GAGTCCTTCCACGATAC.

ROS assay

The reactive oxygen species (ROS) was detected using a ROS assay kit (BestBio, China). According to the manufacturer's protocol, cells were incubated with DCFH-DA (10 µM) for 30 min at 37 °C. The fluorescence intensity of ROS was detected with a Bio-Tek fluorescence microplate reader (Nanjing Detie Experimental Equipment Co., Ltd, China).

qRT-PCR

Total RNA was extracted from the tissues and cells by the Trizol reagent (Invitrogen, USA). A Reverse Transcription Kit (Takara, China) was used to convert RNA into cDNA. The experiment was conducted using SYBRTM Green PCR Master Mix (Thermo Fisher, USA) on a StepOnePlus Real-Time PCR System (Analytikjena, Germany), and the data were calculated using $2^{-\Delta\Delta CT}$. Expression for p53, p21, UCP1, and PGC-1α was normalized to that of GAPDH. Primers were listed: p21: F-CCTGGTGATGTCCGACCTG, R-CCATGAGCGCATCGCAATC; p53: F-CCCCTGTCATCTTTTGTCCCT, R-AGCTGGCAGAATAGCTTATTGAG; UCP1: F-GTGAACCCGACAACCTTCCGAA, R-TGCCAGGCAAGCTGAAACTC; PGC-1α: F-TATGGA GTGACATAGAGTGTGCT, R-GTCGCTACACCACTTCAATCC; GAPDH: F-AGGTCGGTGTGAACGGATTGTG, R-GGGGTCGTTGATGGCAACA; The exosomal miRNA was extracted with a miRNA extraction kit (Vazyme, China). Small nuclear RNA (snRNA) U6 was used as an endogenous control to normalize miRNA expression levels. Primers were listed: miR-152-3p: F-AGGGTCAGTGCATGACAGA, R-TACCAACCAACCCACTCACT; U6: F-CTCGCTTCG GCAGCACA, R-AACGCTTCACGAATTTGCGT.

Western blot

Protein was extracted using ProteoPrep® Total Extraction Sample Kit, and protein concentration was quantified

using BCA kit (Beyotime, China), and transferred onto PVDF membrane using SDS-PAGE. The BSA-blocking solution (5%) was added to the membranes for 2 h of cultivation to prevent protein loss. The membranes were incubated with primary antibodies against p53(60,283–2-Ig), p21(ab109199), UCP1(23,673–1-AP), PGC-1 α (ab106814), and GAPDH(ab245357) at 4 °C overnight. After PBS washing, secondary antibodies were added and incubated for 2 h. Signals were visualized with an enhanced chemiluminescence (ECL; Millipore) kit, and gray values of bands were analyzed.

Cell senescence assay

3 T3-L1 cells were incubated with young exosomes or old exosomes separately. After washing with PBS, the cells were fixed with 4% paraformaldehyde for 15 min at room temperature. The cells were subsequently stained with beta-galactosidase staining solution, then sealed with parafilm and incubated overnight at 37 °C. The senescent cells were visualized by light microscopy.

ELISA assay

IL-6 and TNF- α levels were detected using an IL-6 or TNF- α ELISA kit (Beyotime). The 3 T3-L1 were cultured with young exosomes or old exosomes. Forty eight h post-incubation, the cell-free culture medium was collected and analyzed using the ELISA Kit according to the manufacturer's instructions. The optical density at 450 nm was determined using a microplate reader.

Identification and validation of miR-152-3p target genes

The possible target genes of miR-152-3p were predicted using TargetScan and the miRbase database. The intersected genes that have a p value of 0.05 or higher were selected for further analysis as potential target genes.

Luciferase reporter analysis

The luciferase reporter assay was used to verify the targeting relationship between PGC-1 α and miR-152-3p. In short, the target fragments of wild-type (WT) PGC-1 α and mutant (MUT) PGC-1 α were constructed. Afterward, PGC-1 α -WT or PGC-1 α -MUT were co-transfected with miR-152-3p mimics, miR-152-3p inhibitors or negative controls into 3 T3-L1 cells. After 24 h of transfection, the luciferase activity was determined according to the manufacturer's instructions.

Statistical analysis

Statistical analyses were conducted using GraphPad Prism 8, and the results were reported as the mean $\pm$ standard deviation. Paired t test and one-way ANOVA were used to compare differences between two or more groups. Statistics with $P < 0.05$ were considered significant.

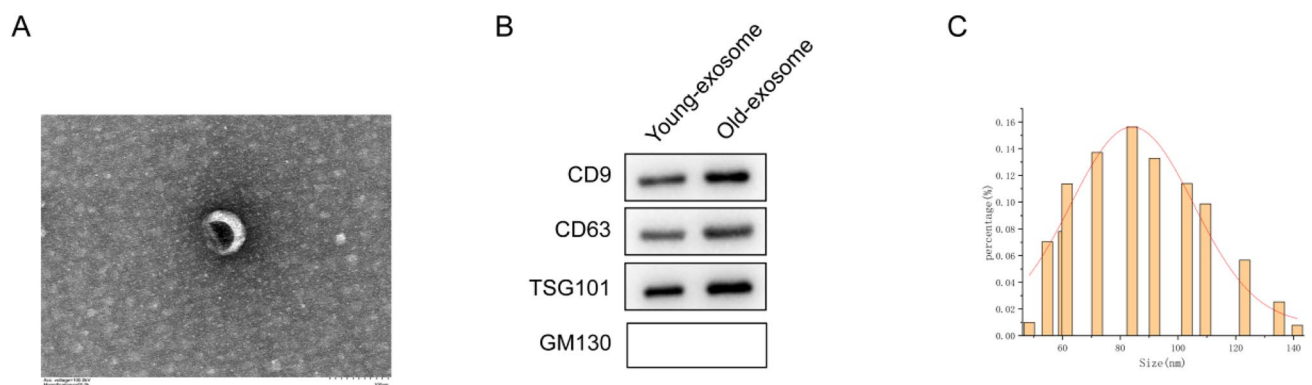


Fig. 1 Identification of exosomes derived from small intestinal epithelial cells. **A** TEM images indicating the typical cup-shaped morphology of exosomes. **B** WB analysis showing the abundant expression of CD9, CD63, TSG101, and the absent expression of GM130 in

young and old exosomes. young-exosome, exosomes extracted from IECs of young mice; old-exosome, exosomes extracted from IECs of old mice. **C** NTA analysis showing the particle size distribution among 60–120 nm

Results

Characterization of exosomes derived from the small intestinal epithelium

As shown in Fig. 1A, the electron microscope displayed the morphology of exosomes extracted from IECs, presenting membrane structure vesicles. Subsequent WB analysis showed that IECs-exos were positive for the characteristic exosomal surface markers including CD9, CD63, and TSG10, and negative for GM130 (Fig. 1B). The results of NTA analysis showed that the diameters of IECs-exos were primarily distributed among 60–120 nm (Fig. 1C). As a result, the vesicles isolated from the IECs were identified as exosomes.

Effect of IECs-exos on 3 T3-L1 cells

To investigate whether IECs-exos affected adipogenesis, 3 T3-L1 preadipocytes were differentiated into adipocytes

by the addition of an adipogenic cocktail containing dexamethasone, IBMX, and insulin. Oil red O staining revealed the accumulation of intracellular lipid droplets, indicating adipogenic differentiation (Fig. 2A).

After preadipocyte differentiation, we studied the role of IECs-exos in differentiated adipocytes. The 3 T3-L1 viability increased when co-cultured with IECs-Exo in a dose-dependent way (Fig. 2B). The IECs-Exo at the concentration of 10^7 particles/mL was used for subsequent experiments. Three subgroups were further divided (control, young-exos, and old-exos). It was discovered that the mtDNA content was the highest in the young-exos group compared with the control group and old-exos group (Fig. 2C). There is a strong correlation between mitochondrial functions and mtDNA integrity, which is partly dependent upon the mitochondrial copy numbers (Gaziev et al. 2014). Similarly, MitoTracker Red showed that the mitochondrial content of young-exos-treated adipocytes was significantly higher than that of the untreated control group and old-exos group (Fig. 2D, F). Moreover, results showed that the level of ROS decreased significantly in the young-exos group (Fig. 2E).

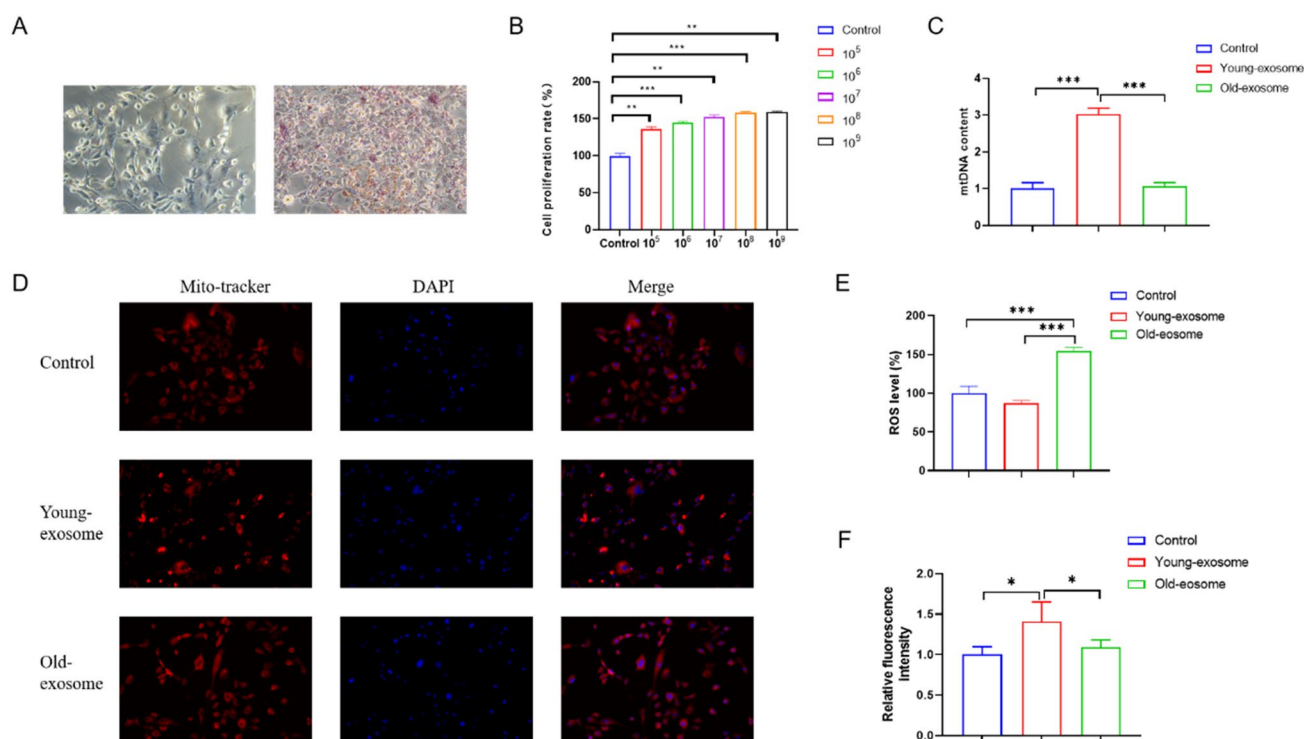


Fig. 2 Exosomes derived from small intestinal cells affect mitochondrial function. **A** Images for Oil red O staining demonstrating lipid droplet accumulation following adipogenic differentiation in 3 T3-L1 cells. 3 T3-L1 preadipocytes (left) were subjected to adipocyte differentiation (right). **B** CCK8 assay showing the concentration-dependent increase on the viability of 3 T3-L1 cells after 24 h of incubation with exosomes. **C** After co-culture with young or old exosomes, real-time PCR showing the most significantly increased relative mtDNA content in young exosomes compared with control or old exosomes. **D**

Images for MitoTracker Red CMXRos (red). Nuclei were stained with DAPI (blue). **E** Statistical results of DCFH-DA staining showing the decreased ROS content in young exosomes compared with control or old exosomes. **F** Statistical results of MitoTracker Red demonstrating the increased mitochondrial activity in young exosomes compared with control or old exosomes. Each sample was measured multiple times in technical duplicate and biologic duplicate ($n = 4$ for each group). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$

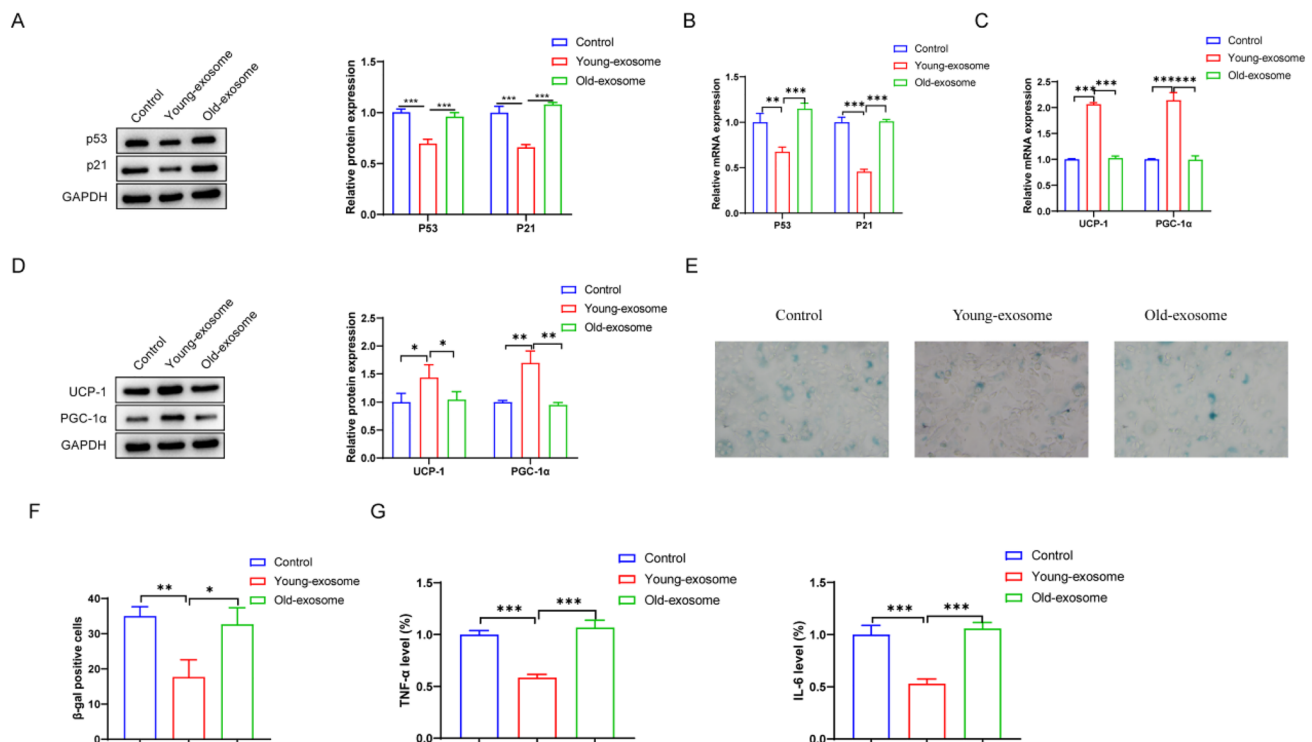


Fig. 3 Effect of exosomes derived from small intestinal cells on the senescence of 3T3L1 cells. After co-culture with young or old exosomes, the specific senescence markers were examined. **A** Quantification of WB analysis for p21 and p53, showing the decreased p21 and p53 expression in young exosomes compared with control or old exosomes, normalized to GAPDH. **B** The relative mRNA expression of p21 and p53 in 3T3L1 cells were detected by qRT-PCR, showing the decreased p21 and p53 expression in young exosomes compared with control or old exosomes. **C** UCP1 and PGC-1α expression were evaluated by qRT-PCR, showing the increased UCP1 and PGC-1α expression in young exosomes compared with control or old exosomes. **D** UCP1, PGC-1α proteins were analyzed by WB, show-

ing the increased UCP1 and PGC-1α expression in young exosomes compared with control or old exosomes, normalized to GAPDH. **E** Images for β-galactosidase staining of 3T3L1 cells. **F** Statistical results of β-galactosidase staining demonstrating the reduced relative numbers of β-galactosidase-positive cells in young exosomes compared with control or old exosomes. **G** The levels of TNF-α and IL-6 were detected by ELISA, showing the declined TNF-α and IL-6 levels in young exosomes compared with control or old exosomes. Each sample was measured multiple times in technical duplicate and biologic duplicate ($n = 4$ for each group). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$

Understanding the molecular mechanisms of ROS accumulation and finding therapies to modulate ROS concentrations are critical to addressing aging and aging-related disease (Houry et al. 2020). Taken together, the young exosomes significantly increased the mitochondrial activity of adipocytes.

Effect of IECs-exos on the senescence of 3T3-L1 cells

In Fig. 3A and B, a reduction was shown in both the mRNA and protein levels of p21 and p53 in young-exosomes-exposed 3T3-L1 cells. However, there was only little difference in the p21 and p53 between the control group and old-exos group. The production of ROS is regulated by uncoupling protein-1 (UCP1), which is the main marker of beige/brown adipocytes (Ost et al. 2017; Cannon et al. 1982; Lin and Lin 1980). PGC-1α is a transcription factor which controls thermogenic genes including the expression of UCP1 (Puigserver et al. 1998). The results showed that both mRNA and protein levels

of UCP1 and PGC-1α were significantly increased in the young-exos group (Fig. 3C–D). A beta-galactosidase staining was used to determine the senescence of cells. At the same time, co-culture with young-exos effectively reduced the number of β-galactosidase-positive cells, compared to the normal control group and the old-exos group (Fig. 3E). Besides, the levels of senescence-associated secretory phenotype (SASP) proinflammatory factors TNF-α and IL-6 showed a significant decrease in the young-exos group, while there were no significant differences between the other groups (Fig. 3F). The aforementioned findings indicate that the related processes of adipocyte aging were influenced by co-culture with young-exos.

miR-152-3p affects the senescence of adipocytes by targeting PGC-1α

To further investigate the mechanism of IECs-exos on adipocytes, TargetScan, and MiRBase were utilized to predict

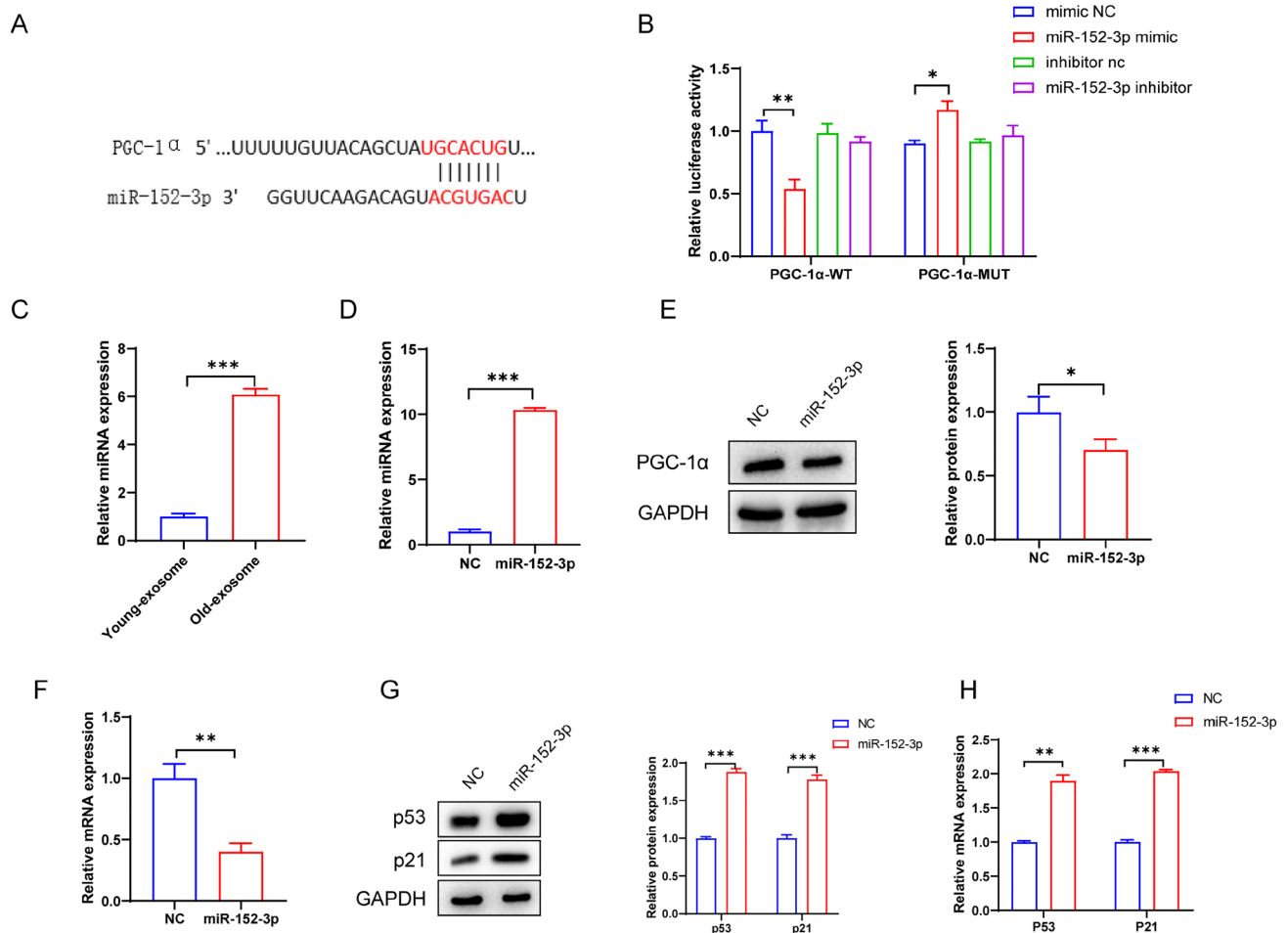


Fig. 4 miR-152-3p targets PGC-1 α and affects the senescence of adipocytes. **A** The possible binding sequences between miR-152-3p and PGC-1 α . **B** Dual luciferase reporter assay was conducted to determine the targeting relationship between PGC-1 α and miR-152-3p, showing the decreased luciferase activity of PGC-1 α -WT and the increased luciferase activity of PGC-1 α -WT after transfection of miR-152-3p mimic. **C** qRT-PCR showing the low relative miR-152-3p expression in young exosomes compared with old exosomes. **D** qRT-PCR showing the significantly increased miR-152-3p expression in the adipocyte cells after transfection with miR-152-3p mimic. **E** WB

analysis showing the declined PGC-1 α protein expression after transfection with miR-152-3p mimic. **F** qRT-PCR showing the declined PGC-1 α mRNA expression after transfection with miR-152-3p mimic. **G** Protein expression of p21 and p53 analyzed by WB, and the statistical results demonstrating the increased p21 and p53 protein expression in adipocytes transfected with miR-152-3p mimic. **H** p21 and p53 expression were evaluated using qRT-PCR, demonstrating the increased p21 and p53 mRNA expression in adipocytes transfected with miR-152-3p mimic. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$

the potential target genes of PGC-1 α . The data showed that PGC-1 α contained a potential conserved binding site for miR-152-3p (Fig. 4A). Meanwhile, it revealed that the miR-152-3p mimics and inhibitor could decrease and increase the luciferase activity of luciferase reporter containing wide type PGC-1 α sequence, while exerting no significant effect on the luciferase activity of mutated PGC-1 α sequence (Fig. 4B). To further confirm the regulation of miR-152-3p on PGC-1 α , the expression of miR-152-3p in IECs-exos was investigated. The result showed that miR-152-3p was highly expressed in exosomes derived from IECs of 18-month mice (Fig. 4C). After transfection with miR-152-3p mimic, miR-152-3p expression was expectedly upregulated (Fig. 4D).

Results of WB and qRT-PCR displayed showed a lower expression of PGC-1 α after the transfection with miR-152-3p mimic (Fig. 4E–F). In parallel, the p21 and p53 were upregulated in cells when miR-152-3p was upregulated (Fig. 4G–H). Taken together, these results imply an essential role for miR-152-3p in the aging process of adipocytes through PGC-1 α .

Effects of miR-152-3p on visceral fat and aging of mice

After animal sacrifice, mouse adipose tissues and kidney were harvested. Notably, the mice in miR-152-3p inhibitor

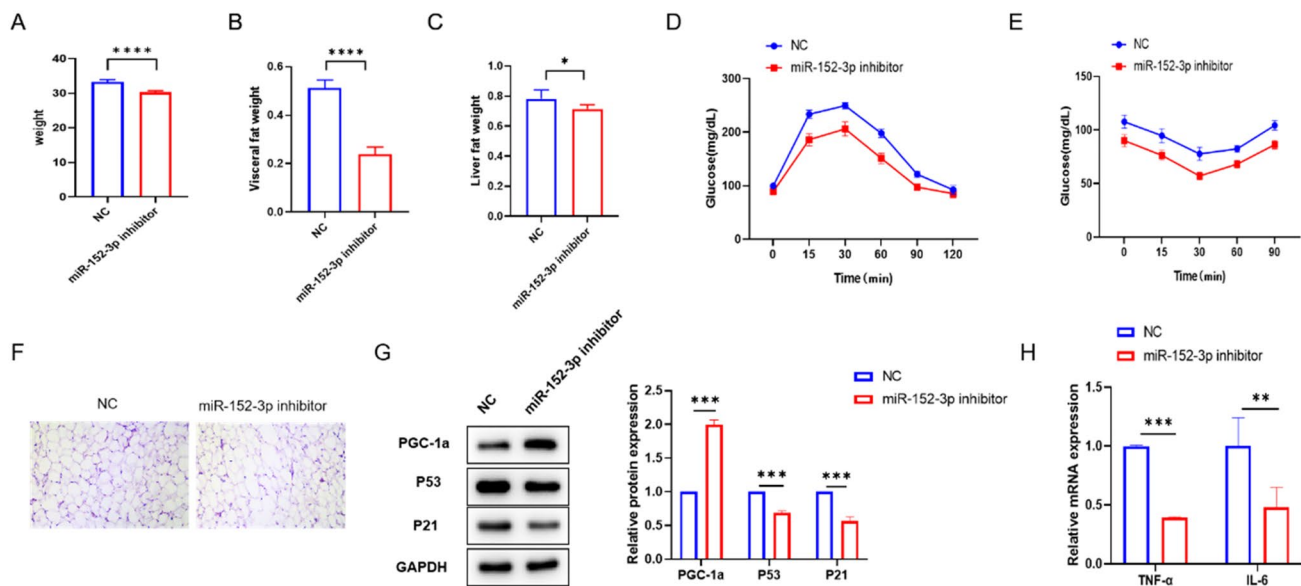


Fig. 5 Effects of miR-152-3p on visceral fat and aging of the mice. **A–C** Statistical results of body weight, adipose tissue and kidney weight of mice, showing the reduced body weight, visceral fat weight and kidney weight after transfection of miR-152-3p inhibitor. **D–E** Insulin sensitivity and glucose tolerance was assessed by ITT and GTT in mice, showing that miR-152-3p inhibitor improved insulin sensitivity and glucose tolerance. **F** Images for H&E staining of adipose tissue sections demonstrating the decreased fat accu-

mulation after transfection of miR-152-3p inhibitor. **G** Protein levels of PGC-1α, p53 and p21 using WB, showing the increased PGC-1α expression, and the declined p53 and p21 expression after transfection of miR-152-3p inhibitor. **H** qRT-PCR analysis of the expression of pro-inflammatory cytokines TNF-α and IL-6, showing the reduced TNF-α and IL-6 expression after transfection of miR-152-3p inhibitor. * $P < 0.05$, *** $P < 0.001$

group had smaller total adipose tissue, decreased body weight and kidney weight compared to NC group mice (Fig. 5B–C). In addition, the results indicated that both insulin sensitivity (Fig. 5D) and glucose tolerance (Fig. 5E) were improved in the miR-152-3p inhibitor group. In addition, adipocyte number was decreased, and more stromovascular cells were present in mice injected with miR-152-3p inhibitor (Fig. 5F). Furthermore, we found that miR-152-3p inhibitor significantly increased the protein levels of PGC-1α, and inhibited the protein levels of p53 and p21 in tissues compared with NC group (Fig. 5G). Finally, the miR-152-3p inhibitor treatment further inhibited the expression of pro-inflammatory cytokines TNF-α and IL-6 in the adipose tissue around the kidneys (Fig. 5H).

Discussion

Our study revealed for the first time that the aging of adipocytes is closely associated with the release of exosomes. In this study, we found that the viability of 3T3-L1 was increased and the degree of aging was inhibited when cells were exposed to exosomes of small intestinal epithelial cells. Changes in age-related features included SA-β gal activity, ROS activation, and SASP proinflammatory factors. The expression levels of miR-152-3p in the exosomes of small

intestinal epithelial cells in young mice was distinguished from that in old mice, which affected the expression of PGC-1α and was related to the aging of adipocytes.

PGC-1α is the main regulatory factor for mitochondrial biogenesis (Barbatelli et al. 2010). As the central regulatory factor for heat production in brown fat, PGC-1 α is mainly expressed in BAT, regulating the entire heat or energy production process (Lin et al. 2005; Uldry et al. 2006). BAT is mainly located in the interscapular space of mice, as well as in the interscapular, supraclavicular, suprarenal, and paraaortic spaces of humans (Cinti 2019). The rodent study has shown that thermogenic adipocytes influence adipose tissue biology through anti-inflammatory, anti-fibrotic, and angiogenic effects. In addition to the thermogenic effect, it also affects insulin sensitivity and secretes other small molecules such as extracellular vesicles containing miRNA, which play a role in systemic insulin sensitivity and energy homeostasis (Silva and Amato 2022). As age increases, overall changes in the quality, distribution, and function of adipose tissues have a significant impact on other tissues (Palmer and Kirkland 2016). Previous studies have shown that thermogenic fats have functional activity in adults, and their presence is related to body weight and metabolic health indicators (Cypess et al. 2009). Strategies aimed at improving health and even extending lifespan in adipose tissues have

achieved encouraging results (Ou et al. 2022). Although most studies have focused on white adipose tissues, recent evidence suggests that thermogenic adipocytes may also be used to improve age-related diseases and even extend lifespan (Zoico et al. 2019). Therefore, the relationship between adipocytes and aging is worthy of attention. Our research indicated that the exosomes of the small intestine affected the visceral fat function and aging, and whether they affect the body's aging and potential mechanisms needs further research.

Recently, exosomes have been proven as beneficial components in the blood to prolong mouse lifespan, indicating that some specific exosomes may have anti-aging functions (Luo et al. 2021). Inhibiting the release of extracellular vesicles accelerates aging both *in vitro* and *in vivo* (Zou et al. 2023). Importantly, the exosomes derived from multiple tissues have been reported to participate in the aging process. For example, exosomes derived from stem cells or human dermal fibroblasts or human umbilical vein endothelial cells can ameliorate skin photoaging (Hajialiasgari Najafabadi et al. 2024). In particular, exosomes from adipose-derived stem cells attenuate adipose inflammation and obesity (Zhao et al. 2018). However, the protective role of exosomes derived from small intestinal endocrine cells has only been investigated in islets β cells (Yang et al. 2024). In this study, we found that exosomes derived from small intestinal epithelial cells altered the aging process of adipocytes, for the first time revealing the pivotal role of the exosomes derived from intestine in the context of the aging of adipocytes. Besides, exosomes can transfer miR-152-3p to mediate the browning functions of adipocytes. MiR-152-3p belongs to the highly conserved microRNA and previous study has focused on the regulation of miR-152-3p in tumor cell proliferation, differentiation, and survival (Ma et al. 2020). Yuan Fan et al. found that miR-152 regulates 3T3-L1 cell differentiation and may have therapeutic significance for obesity and obesity-related metabolic syndrome (Fan et al. 2025). According to reports, miR-152 is significantly upregulated during the replication aging process of human lung fibroblasts and human skin fibroblasts, and significantly upregulated during the aging process of human dental pulp stem cells (Mancini et al. 2012; Dou et al. 2023). MiR-152-3p shares the same seed sequence of approximately six-seven nucleotides with other members of the miR-152 family, which has been reported to represent the blood miR characteristics of segmental aging in Down syndrome patients (Wang et al. 2016; Morsiani et al. 2022). Here, it was found that the overexpression of miR-152-3p induced the expression of aging markers in adipocytes, suggesting that it might be a potential mechanism of action of exosomes. The study which introduced the aging mechanism of the exosomes derived from small intestines in adipose tissues might provide promising targets and strategies for improving health.

As a limitation of experiments *in vivo*, the targeting relationship between IECs-derived exosomal miR-152-3p and PGC-1 α has not been further validated in animal experiments. Besides, only murine 3T3-L1 cells were used here and the functions on primary human adipocytes have not been determined.

In summary, we demonstrated that exosomes isolated from small intestinal epithelial cells in young mice enhanced the viability and inhibited the aging of adipocytes. Second, exosomes may act on PGC-1 α through miR-152-3p, regulating visceral fat browning, fat aging and SASPs expression. MiR-152-3p should be considered as a potential therapeutic molecule for the aging of adipocytes. However, further study is needed to investigate the detailed mechanism underlying the exosome effects of adipocytes.

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Data availability The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare that they have no competing interests.

Consent for publication Not applicable.

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